Stability analysis of Arbitrary-Lagrangian-Eulerian ADER-DG methods on classical and degenerate spacetime geometries

Mauro Bonafini[a,*], Davide Torlo[b], Elena Gaburro[a]

[a]*Department of Computer Science, University of Verona, Strada le Grazie 15, Verona, 37134, Italy*
[b]*Mathematics Department "Guido Castelnuovo", University of Rome Sapienza, p.le Aldo Moro, 5, Rome, 00189, Italy*

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Abstract

In this paper, we present a thorough von Neumann stability analysis of explicit and implicit Arbitrary-Lagrangian-Eulerian (ALE) ADER discontinuous Galerkin (DG) methods on *classical* and *degenerate* spacetime *geometries* for hyperbolic equations.

First, we rigorously study CFL stability conditions for the *explicit* ADER-DG method, confirming results widely used in the literature while specifying their limitations. Moreover, we discuss stability bounds for ALE methods and characterise the admissible range of grid velocities once a target CFL is fixed.

Next, we extend the stability study to ADER-DG in the presence of *degenerate spacetime elements*, with *zero size* at the beginning and the end of the time step, but with a *non zero spacetime volume*. This kind of elements has been introduced in a series of articles on direct ALE methods by Gaburro *et al.* to connect via spacetime control volumes regenerated Voronoi tessellations after a topology change. Here, we imitate this behaviour in a 1d surrogate setting by fictitiously inserting degenerate elements in between two cells. We show that over this simplified degenerate spacetime geometry, both for the explicit and implicit ADER-DG, the von Neumann analysis leads to the *same* CFL *stability conditions* as those for classical geometries, laying the theoretical foundations for their use in the context of ALE methods.

Keywords: von Neumann stability analysis, CFL stability constraints, Arbitrary-Lagrangian-Eulerian methods, ADER discontinuous Galerkin methods, explicit and implicit high order methods, degenerate spacetime geometries

1. Introduction

The spacetime predictor-corrector ADER approach, introduced in its original formulation in [54, 73], extended to nonlinear systems in [74, 72] and to unstructured domains in [41, 40, 19, 10], and finally presented with a modern and effective approach in [16], is nowadays widely used to reach high order of accuracy also in time when solving hyperbolic partial differential equations (PDEs). It consists of an element-local iterative method to construct a piecewise polynomial, called predictor, that represents an approximation of high order of accuracy both in space and time of the PDE solution. Then, in the corrector phase, the predictor is directly inserted into a one-step update formula to evolve the solution from one time step to the next one. The method has been shown to be very reliable and effective since it introduces minimal dissipation and can be parallelized with minimal communication overhead [25, 18].

In particular, ADER approaches can be used within the framework of discontinuous Galerkin (DG) finite element methods, as originally proposed in [64, 20, 46, 35]. Later, the ADER-DG methods have been applied to many different hyperbolic systems, starting from the classical Euler and shallow water equations [65, 26, 68, 14], multiphase models [68, 42], magnetohydrodynamics [2, 24], dispersive, turbulent and reactive flows [9, 78, 8, 59], up to the study of much more complex systems, as unified models for continuum mechanics [21, 22, 71, 7, 13, 43] and various first order reformulations of the Einstein field equations for general relativity [23, 79, 55].

*Corresponding author
Email addresses: mauro.bonafini@univr.it (Mauro Bonafini), davide.torlo@uniroma1.it (Davide Torlo), elena.gaburro@univr.it (Elena Gaburro)

Over the years, many authors have worked with ADER-DG schemes proposing performance improvements [36, 51, 75, 60, 48], implicit [17] and semi-implicit [57, 6, 62] formulations, and in particular their use in the context of *direct* Arbitrary-Lagrangian-Eulerian (ALE) methods [5]. In this latter framework, the mesh moves following as close as possible the local fluid flow and, in order to evolve the PDE from one mesh to the next one, an effective approach consists in connecting them via spacetime control volumes, possibly degenerate [30, 31, 29, 33], over which integrating the PDE. Here, ADER-DG methods offer the ideal tools to build a one-step solver that is automatically able to integrate the PDE in space and time and to perform *directly* the time update (without any need for projection and reconstruction procedures, typical of *indirect* approaches [47, 3]).

Given the widespread adoption of ADER and ADER-DG methods, a rigorous theoretical investigation of their fundamental properties becomes necessary. While the convergence of the local spacetime predictor has been established for linear problems in [39] and extended to nonlinear systems in [7], stability studies still require further investigation. Regarding stability, some authors have addressed ADER-type schemes in the simplified framework of ordinary differential equations [37, 62, 61], but a comprehensive analysis in the PDE setting is still missing. The seminal work [16] provides stability considerations only for low polynomial degrees, while for higher orders the admissible CFL numbers are typically determined empirically. Moreover, to the best of our knowledge, no dedicated analysis has been carried out for the newly introduced ALE ADER-DG methods on possibly degenerate spacetime geometries.

For the above reasons, it is of general interest to establish a rigorous stability analysis of ADER-DG methods. There are several strategies to study the stability of numerical methods, some of which apply only to linear problems, while others are also able to treat nonlinear problems. For linear problems, one can look at the Lax–Richtmyer stability [44], which, however, turns out to be difficult to prove in many situations. In the context of periodic domains, the von Neumann stability analysis guarantees Lax–Richtmyer stability and can be easily performed using Fourier modes instead of fully discretized problems, strongly reducing the complexity of the stability analysis. Indeed, there are many works that have studied the von Neumann stability to deduce CFL conditions or other parameter conditions, e.g. for residual distribution in 1d and 2d [52, 53] and for DG in 1d [70].

The extension of linear analyses to nonlinear problems is not straightforward, but the results obtained in the linear setting are often a good indicator of the CFL that should be used in the presence of non linearities. A direct study of the L^2 stability can be done also in the nonlinear case, but it would still require an analysis of all the degrees of freedom and it is typically easy to apply only to simple time discretizations (explicit/implicit Euler, SSPRK) [15, 45]. Another way to tackle the stability would be to mimic the integration by parts at the discrete level to guarantee energy or entropy stability in the spatial discretization [27, 56], and also to apply relaxation in time to preserve or dissipate a global entropy [66, 1, 34].

1.1. Aim and structure of this paper

The scope of this paper is to fill the theoretical gap in ADER-DG methods by providing a thorough von Neumann stability analysis, accounting for both the explicit and implicit formulations on fixed or moving meshes. In Section 2, to fix the notation and to make the paper self-contained, we present the ADER-DG scheme written in the so-called direct Arbitrary-Lagrangian-Eulerian framework. We apply it to scalar nonlinear hyperbolic PDEs on a classical 1d+time geometry. Next, in Section 3, we perform the von Neumann stability analysis of the method and show numerical results for its convergence order. Then, in Section 4, we introduce into our computational domain *fictitious spacetime degenerate elements* that mimic the behaviour of the hole-like sliver elements used in [30, 28, 31, 29] to connect meshes with topology changes, and we study the stability and convergence of the method in their presence. Finally, in Section 5, we draw our conclusions and we provide an outlook on future developments.

2. Explicit and implicit formulation of ALE ADER-DG methods

Let us consider a first order nonlinear scalar hyperbolic PDE

$$\partial_t Q(x,t) + \partial_x f(Q(x,t)) = 0, \tag{1}$$

where Q denotes the conserved scalar variable and f denotes the flux function.

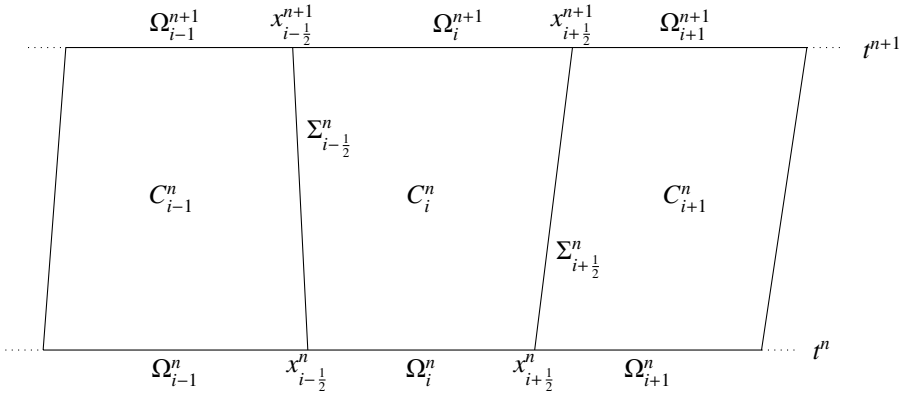


Figure 1: Prototype configuration of a classical spacetime domain discretization. At each time t^n the spatial domain is subdivided into elements Ω^n_i, connected in spacetime by control volumes C^n_i. Each control volume C^n_i neighbours the previous and next control volume through the lateral interfaces $\Sigma^n_{i-1/2}$ and $\Sigma^n_{i+1/2}$ respectively.

Geometry description.
Given an initial domain $\Omega^0 = [x_L, x_R]$, with $x_L, x_R \in \mathbb{R}$, we consider N_e+1 points $x_L = x^0_{1/2} < x^0_{3/2} < \cdots < x^0_{N_e+1/2} = x_R$ and define N_e spatial elements $\Omega^0_i := [x^0_{i-1/2}, x^0_{i+1/2}]$, $i = 1, \dots, N_e$. Given an initial time $t^0 \in \mathbb{R}$, we discretize time by means of intervals $[t^n, t^{n+1}]$ with $n \in \mathbb{N}$ of step size $\Delta t^n = t^{n+1} - t^n > 0$ for $n \geq 0$. When not ambiguous, we will simply use the notation Δt for Δt^n. In a Lagrangian perspective, we also assume the spatial computational domain to depend on t: at each time t^n our computational domain is then denoted as $\Omega^n = [x^n_L, x^n_R]$, and we consider again $N_e + 1$ points $x^n_L = x^n_{1/2} < x^n_{3/2} < \cdots < x^n_{N_e+1/2} = x^n_R$ and define N_e elements $\Omega^n_i := [x^n_{i-1/2}, x^n_{i+1/2}]$, $i = 1, \dots, N_e$. We denote the length of each element by $\Delta x^n_i := |\Omega^n_i| = (x^n_{i+1/2} - x^n_{i-1/2})$ and define its barycentre $b^n_i := (x^n_{i+1/2} + x^n_{i-1/2})/2$. For each $i = 1, \dots, N_e$, we also define $\Sigma^n_{i\pm1/2}$ as the segment connecting $(x^n_{i\pm1/2}, t^n)$ to $(x^{n+1}_{i\pm1/2}, t^{n+1})$ and introduce the spacetime control volume C^n_i as the polygonal set with vertices $(x^n_{i-1/2}, t^n), (x^n_{i+1/2}, t^n), (x^{n+1}_{i+1/2}, t^{n+1}), (x^{n+1}_{i-1/2}, t^{n+1})$. In particular, $\partial C^n_i = \Sigma^n_{i-1/2} \cup (\Omega^n_i \times \{t^n\}) \cup \Sigma^n_{i+1/2} \cup (\Omega^{n+1}_i \times \{t^{n+1}\})$. We refer to Figure 1 for a visual depiction of the setting.

Families of basis functions.
We fix a polynomial degree $N \geq 0$. For each $n \geq 0$ and $i \in \{1, \dots, N_e\}$, we introduce three different sets of basis functions. First, we define over each domain Ω^n_i the set of *spatial* basis functions $\{\phi^n_{i,\ell}\}_\ell$. These are modal basis functions centred at the cell barycentre and scaled by the element length, defined as

$$\phi^n_{i,\ell} \colon \Omega^n_i \to \mathbb{R},\ x \mapsto \left(\frac{x - b^n_i}{\Delta x^n_i}\right)^\ell \quad \text{for } \ell = 0, \dots, N. \tag{2}$$

Next, we define over each spacetime control volume two different sets of basis functions. First, we define the set of *moving* basis functions $\{\psi^n_{i,\ell}\}_\ell$, obtained by tracking the spatial basis functions along the trajectory of the cell barycentre from t^n to t^{n+1}. They are defined as

$$\begin{aligned} &\psi^n_{i,\ell} \colon C^n_i \to \mathbb{R},\ (x,t) \mapsto \left(\frac{x - \tilde{b}^n_i(t)}{\Delta x^n_i}\right)^\ell \quad \text{for } \ell = 0, \dots, N, \\ &\text{with } \tilde{b}^n_i(t) = \left(1 - \frac{t - t^n}{\Delta t^n}\right) b^n_i + \frac{t - t^n}{\Delta t^n} b^{n+1}_i. \end{aligned} \tag{3}$$

Next, we introduce the set of *spacetime* basis functions $\{\theta^n_{i,\ell}\}_\ell$. These are modal basis functions centred at (b^n_i, t^n) and

scaled by the element spacetime lengths, defined as

$$\begin{aligned}
&\theta^n_{i,\ell}\colon C^n_i \to \mathbb{R},\ (x,t) \mapsto \left(\frac{x-b^n_i}{\Delta x^n_i}\right)^{\ell_1}\left(\frac{t-t^n}{\Delta t^n}\right)^{\ell_2}\\
&\text{for } \ell = \ell_1 + \ell_2(2N+3-\ell_2)/2,\ 0 \le \ell_1+\ell_2 \le N,\\
&\text{so that } \ell = 0,\dots,N_{st}-1, \text{ with } N_{st} := (N+1)(N+2)/2.
\end{aligned} \tag{4}$$

The chosen indexing first considers purely spatial functions and progressively incorporates higher powers in time, getting this way the full set of N_{st} spacetime basis functions up to total degree N. For $N = 4$, up to shifting and scaling, the indexing scheme and the corresponding basis functions are given as follows:

	space index ℓ_1				
time index ℓ_2	0	1	2	3	4
0	$0:1$	$1:x$	$2:x^2$	$3:x^3$	$4:x^4$
1	$5:t$	$6:xt$	$7:x^2t$	$8:x^3t$	
2	$9:t^2$	$10:xt^2$	$11:x^2t^2$		
3	$12:t^3$	$13:xt^3$			
4	$14:t^4$				

Representation of the approximate solution.
The conserved variable Q is represented over each grid element Ω^n_i via a piecewise polynomial function u^n_i in the form

$$u^n_i(x) = \sum_{\ell=0}^{N} \phi^n_{i,\ell}(x)\,\hat{u}^n_{i,\ell} \quad \text{for } x \in \Omega^n_i,\ i \in \{1,\dots,N_e\}, n \ge 0, \tag{5}$$

where $\{\hat{u}^n_{i,\ell}\}_{n,i,\ell} \subset \mathbb{R}$ are the degrees of freedom. We also write $\hat{u}^n_i \in \mathbb{R}^{N+1}$ to identify the corresponding vector of coefficients. The family $\{u^n_i\}_i$ defines a global (discontinuous) approximant u^n over Ω^n, which is well-defined everywhere except at the interfaces between elements. Approximations defined on each control volume C^n_i will usually take the form of a spacetime polynomial function q^n_i written as

$$q^n_i(x,t) = \sum_{\ell=0}^{N_{st}-1} \theta^n_{i,\ell}(x,t)\hat{q}^n_{i,\ell} \quad \text{for } (x,t) \in C^n_i,\ i \in \{1,\dots,N_e\}, n \ge 0, \tag{6}$$

where $\{\hat{q}^n_{i,\ell}\}_{n,i,\ell} \subset \mathbb{R}$ are the degrees of freedom. Again, we write $\hat{q}^n_i \in \mathbb{R}^{N_{st}}$ to identify the corresponding vector of coefficients.

2.1. Explicit ALE ADER-DG method

Given an approximate solution u^n in the form (5), the explicit ALE ADER-DG method is a two steps method aiming to compute the next approximant u^{n+1}. These steps are termed *predictor* and *corrector* step.

2.1.1. Predictor step

The aim of this step is to build a *local* approximation of the governing PDE (1) over each spacetime control volume C^n_i, using u^n_i as initial condition at time t^n. Over a given spacetime control volume C^n_i, $i = 1,\dots,N_e$, we seek a spacetime polynomial function q^n_i of the form (6) whose coefficients $\hat{q}^n_i$ are to be determined so that q^n_i approximates the element-local PDE problem

$$\partial_t q^n_i(x,t) + \partial_x f(q^n_i(x,t)) = 0 \quad \text{over } C^n_i,$$

with u^n_i to be considered as an inflow condition on Ω^n_i. We fix now any spacetime test function $\theta^n_{i,k}$, $k = 0,\dots,N_{st}-1$, multiply the above PDE by $\theta^n_{i,k}$ and integrate over the given control volume C^n_i to get

$$\int_{C^n_i} \theta^n_{i,k}(x,t)\left[\partial_t q^n_i(x,t) + \partial_x f(q^n_i(x,t))\right] \mathrm{d}x\mathrm{d}t = 0 \quad \text{for } k = 0,\dots,N_{st}-1. \tag{7}$$

Then, we rewrite the first term in (7) by taking into account a potential jump of q_i^n at the boundary Ω_i^n of C_i^n via a simplified path-conservative approach [58, 12, 11], obtaining

$$\begin{aligned}&\int_{C_i^n} \theta_{i,k}^n(x,t)\partial_t q_i^n(x,t)\,\mathrm{d}x\mathrm{d}t + \int_{\Omega_i^n} \theta_{i,k}^n(x,t^n)\,\left(q_i^n(x,t^n) - u_i^n(x)\right)\,\mathrm{d}x \\ &+ \int_{C_i^n} \theta_{i,k}^n(x,t)\partial_x f(q_i^n(x,t))\,\mathrm{d}x\mathrm{d}t = 0 \qquad \text{for } k = 0,\dots,N_{st}-1.\end{aligned} \tag{8}$$

For a given q_i^n, we now define the vector $\hat{f}_i^n = (\hat{f}_{i,0}^n, \dots, \hat{f}_{i,N_{st}-1}^n)$ so that

$$f_i^n(x,t) = \sum_{\ell=0}^{N_{st}-1} \theta_{i,\ell}^n(x,t)\hat{f}_{i,\ell}^n \qquad \begin{array}{l}\text{is the } L^2\text{-projection of } f(q_i^n) \text{ over the finite}\\ \text{dimensional space span}\{\theta_{i,0}^n,\dots,\theta_{i,N_{st}-1}^n\}.\end{array} \tag{9}$$

Finally, by replacing in (8) the term $f(q_i^n)$ with f_i^n, we get

$$\begin{aligned}&\int_{C_i^n} \theta_{i,k}^n(x,t)\partial_t q_i^n(x,t)\,\mathrm{d}x\mathrm{d}t + \int_{\Omega_i^n} \theta_{i,k}^n(x,t^n) q_i^n(x,t^n)\,\mathrm{d}x \\ &= \int_{\Omega_i^n} \theta_{i,k}^n(x,t^n) u_i^n(x)\,\mathrm{d}x - \int_{C_i^n} \theta_{i,k}^n(x,t)\partial_x f_i^n(x,t)\,\mathrm{d}x\mathrm{d}t \qquad \text{for } k = 0,\dots,N_{st}-1.\end{aligned} \tag{10}$$

By expanding q_i^n and u_i^n in (10), see (6) and (5), we obtain an implicit system of equations for the unknown coefficient vector $\hat{q}_i^n$. Here, the coefficient vector $\hat{f}_i^n$ is indeed a (generally nonlinear) function of $\hat{q}_i^n$, and so to approximate $\hat{q}_i^n$ we employ a fixed point Picard iteration, as detailed in [16, 38, 7]. We can consider the vector $\hat{q}_i^{n,(0)} := (\hat{u}_{i,0}^n, \dots, \hat{u}_{i,N}^n, 0, \dots, 0)$ as starting point for the iteration and remark that such fixed point procedure has already been proved to be convergent and to yield the desired order of accuracy, see [39, 7, 37] for more details.

Upon convergence, the predictor step provides us with a set of locally defined high order polynomials q_i^n over each control volume C_i^n. These polynomials will serve as approximants of the solution in the interior of C_i^n and will be used in the computation of the numerical fluxes at the interfaces.

2.1.2. Corrector step

For each $i \in \{1,\dots,N_e\}$, we multiply the governing equation (1) by a moving basis function $\psi_{i,k}^n$, $k = 0,\dots,N$, and integrate over the control volume C_i^n to obtain

$$\int_{C_i^n} \psi_{i,k}^n(x,t)\left[\partial_t Q(x,t) + \partial_x f(Q(x,t))\right]\,\mathrm{d}x\mathrm{d}t = 0 \quad \text{for } k = 0,\dots,N.$$

Next, by integration by parts, we get

$$\int_{\partial C_i^n} \psi_{i,k}^n (Q\,\hat{n}_{i,t}^n + f(Q)\,\hat{n}_{i,x}^n)\,\mathrm{d}s - \int_{C_i^n} \left[\partial_t \psi_{i,k}^n Q + \partial_x \psi_{i,k}^n f(Q)\right]\mathrm{d}x\mathrm{d}t = 0 \quad \text{for } k = 0,\dots,N,$$

where $\hat{\mathbf{n}}_i^n = (\hat{n}_{i,x}^n, \hat{n}_{i,t}^n)$ denotes the outward pointing unit normal vector on the spacetime faces composing the boundary ∂C_i^n. We now decompose ∂C_i^n into Ω_i^n, Ω_i^{n+1} and the two lateral faces $\Sigma_{i\pm1/2}^n$, we recall that u_i^n (resp. u_i^{n+1}) approximates Q on Ω_i^n (resp. Ω_i^{n+1}) and that the predictor q_i^n approximates Q inside the control volume C_i^n. Upon introducing a suitable numerical flux function $\mathcal{F}: \mathbb{R}\times\mathbb{R}\times S^1 \to \mathbb{R}$, we obtain

$$\begin{aligned}&\int_{\Omega_i^{n+1}} \psi_{i,k}^n(x,t^{n+1}) u_i^{n+1}(x)\,\mathrm{d}x = \int_{\Omega_i^n} \psi_{i,k}^n(x,t^n) u_i^n(x)\,\mathrm{d}x - \int_{\Sigma_{i-\frac{1}{2}}^n} \psi_{i,k}^n \mathcal{F}(q_i^n, q_{i-1}^n, \hat{\mathbf{n}}_i^n)\,\mathrm{d}s - \int_{\Sigma_{i+\frac{1}{2}}^n} \psi_{i,k}^n \mathcal{F}(q_i^n, q_{i+1}^n, \hat{\mathbf{n}}_i^n)\,\mathrm{d}s \\ &+ \int_{C_i^n} \partial_t \psi_{i,k}^n(x,t) q_i^n(x,t)\,\mathrm{d}x\mathrm{d}t + \int_{C_i^n} \partial_x \psi_{i,k}^n(x,t) f(q_i^n(x,t))\,\mathrm{d}x\mathrm{d}t \quad \text{for } k = 0,\dots,N,\end{aligned} \tag{11}$$

where the coefficient vector $\hat{u}_i^{n+1}$ can be computed *explicitly* upon knowing $\hat{u}_i^n$ and by integrating the remaining terms that depend only on the already computed predictors q_i^n. The above formula is termed the *corrector* step of the ADER-DG scheme.

The numerical flux function $\mathcal{F}$ is computed via an ALE Riemann solver applied to the inner and outer boundary-extrapolated data q^- and q^+ at each boundary. Here, the simplest choice consists in adopting a Rusanov-type [69] ALE flux. For $q^-, q^+ \in \mathbb{R}$ the interior and exterior extrapolated values, and for an outgoing normal vector $\hat{\mathbf{n}} = (\hat{n}_x, \hat{n}_t) \in S^1$, we set

$$\mathcal{F}(q^-, q^+, \hat{\mathbf{n}}) = \frac{1}{2}\left(f(q^+) + f(q^-)\right)\hat{n}_x + \frac{1}{2}\left(q^+ + q^-\right)\hat{n}_t - \frac{1}{2}s_{\max}\left(q^+ - q^-\right), \tag{12}$$

where, in full generality, $s_{\max}$ is the maximum of the spectral radii of the ALE Jacobian matrix w.r.t. the normal direction in space evaluated at q^- and q^+, which here simplifies to

$$s_{\max} = \max\{|f'(q^-) - v|, |f'(q^+) - v|\} \cdot |\hat{n}_x|,$$

where $v = -\hat{n}_t/|\hat{n}_x|$ is the local grid velocity. For the general expression for systems and in higher dimension we refer to [4].

2.2. Implicit ALE ADER-DG method

Opposite to the explicit method, the implicit ALE ADER-DG method directly seeks a *global* approximant q^n by means of simultaneously computing all approximants q_i^n over each control volume C_i^n. For each $i \in \{1, \dots, N_e\}$, we multiply the governing equation (1) by a spacetime basis function $\theta_{i,k}^n$, $k = 0, \dots, N_{st} - 1$, and integrate over the control volume C_i^n to obtain

$$\int_{C_i^n} \theta_{i,k}^n(x,t)\left[\partial_t Q(x,t) + \partial_x f(Q(x,t))\right] \mathrm{d}x\mathrm{d}t = 0 \quad \text{for } k = 0, \dots, N_{st} - 1.$$

Next, by integration by parts, we get

$$\int_{\partial C_i^n} \theta_{i,k}^n(f(Q), Q) \cdot \hat{\mathbf{n}}_i^n \, \mathrm{d}s - \int_{C_i^n} \left[\partial_t \theta_{i,k}^n Q + \partial_x \theta_{i,k}^n f(Q)\right] \mathrm{d}x\mathrm{d}t = 0 \quad \text{for } k = 0, \dots, N_{st} - 1,$$

where $\hat{\mathbf{n}}_i^n = (\hat{n}_{i,x}^n, \hat{n}_{i,t}^n)$ denotes the outward pointing spacetime unit normal. Now, on each control volume C_i^n we substitute Q by an approximant q_i^n in the form (6), we decompose the boundary integral into four integrals (one over each of the four faces), we introduce a suitable numerical flux $\mathcal{F}$ and require $q_i^n(x, t^n) = u_i^n(x)$ for $x \in \Omega_i^n$, to get

$$\begin{aligned}
&\int_{\Omega_i^{n+1}} \theta_{i,k}^n(x, t^{n+1}) q_i^n(x, t^{n+1}) \, \mathrm{d}x - \int_{C_i^n} \partial_t \theta_{i,k}^n(x,t) q_i^n(x,t) \, \mathrm{d}x\mathrm{d}t - \int_{C_i^n} \partial_x \theta_{i,k}^n(x,t) f_i^n(x,t) \, \mathrm{d}x\mathrm{d}t \\
&+ \int_{\Sigma_{i-\frac{1}{2}}^n} \theta_{i,k}^n \mathcal{F}(q_i^n, q_{i-1}^n, \hat{\mathbf{n}}_i^n) \, \mathrm{d}s + \int_{\Sigma_{i+\frac{1}{2}}^n} \theta_{i,k}^n \mathcal{F}(q_i^n, q_{i+1}^n, \hat{\mathbf{n}}_i^n) \, \mathrm{d}s = \int_{\Omega_i^n} \theta_{i,k}^n(x, t^n) u_i^n(x) \, \mathrm{d}x \\
&\text{for } i = 1, \dots, N_e \text{ and } k = 0, \dots, N_{st} - 1,
\end{aligned} \tag{13}$$

where f_i^n is defined as in (9). The set of equations (13) implicitly defines a generally nonlinear system for the coefficients $\{\hat{q}_{i,\ell}^n\}_{i,\ell}$, which we solve by means of a Newton iteration coupled with a GMRES algorithm. Once each q_i^n has been computed, the next approximant u_i^{n+1} can be recovered through an L^2 projection of $q_i^n(\cdot, t^{n+1})$ with respect to the basis $\{\phi_{i,\ell}^{n+1}\}_\ell$. Note that here, in parallel to the explicit case, we have termed the implicit version of the scheme as implicit ADER-DG, while in other contexts the same method may be referred to as implicit spacetime DG.

3. Stability of ALE ADER-DG methods on classical geometries

We move now to the focus of this paper. We are interested in studying the stability of the explicit ADER-DG method introduced in Section 2 by means of the well known von Neumann stability analysis technique [44, Sections 9.6 and 10.5]. We consider the linear advection equation (LAE) for a constant advection velocity $a > 0$, so that

$f(Q) = aQ$. We restrict ourselves to the domain $\Omega = [0, 2\pi]$ and consider a uniform grid of N_e elements of size $\Delta x > 0$ and a positive Δt. For a given wave number $\kappa \in \mathbb{N}$, we consider the periodic initial condition $u_0(x) = e^{\mathrm{i}\kappa x}$ and close the system with periodic boundary conditions. Since the initial condition satisfies $u_0(x - \Delta x) = e^{-\mathrm{i}\kappa\Delta x} u_0(x)$ for every $x \in \mathbb{R}$, it can be easily proved that the coefficients generated by an ADER-DG method on a fixed grid satisfy $\hat{u}_{i-1}^n = e^{-\mathrm{i}\kappa\Delta x}\hat{u}_i^n$ for any i, n (with the appropriate modifications at the boundaries).

The purpose of this section is then to study some necessary CFL-type bounds for the ADER-DG scheme applied to the periodic LAE. These bounds are required to maintain a bounded norm for the coefficient vectors $\{\hat{u}_i^n\}_i$ over time, and they provide an informed guess for time step selection in complex scenarios where a direct stability analysis is less feasible.

3.1. Stability of the explicit method

To describe the explicit ADER-DG as applied to this setting, we first observe that we can drop the dependency of basis functions on the individual elements, since each Ω_i^n can be obtained (up to translation) from the interval $[0, 1]$ rescaled by Δx and each control volume C_i^n from the square $[0, 1] \times [0, 1]$ rescaled by Δx in space and by Δt in time. We conveniently define the reference basis functions

$$\begin{aligned} \phi_\ell \colon [0,1] \to \mathbb{R},\ \xi \mapsto (\xi - 1/2)^\ell \quad & \text{for } \ell = 0, \dots, N, \\ \psi_\ell \colon [0,1] \times [0,1] \to \mathbb{R},\ (\xi, \tau) \mapsto \phi_\ell(\xi) \quad & \text{for } \ell = 0, \dots, N, \end{aligned}$$

and

$$\theta_\ell \colon [0,1] \times [0,1] \to \mathbb{R},\ (\xi, \tau) \mapsto (\xi - 1/2)^{\ell_1}\ \tau^{\ell_2} \quad \text{for } \ell = \ell_1 + \ell_2(2N + 3 - \ell_2)/2,\ 0 \le \ell_1 + \ell_2 \le N.$$

Since $f(Q) = aQ$, each predictor step in (10) simplifies to

$$\hat{q}_i^n = \left(K_\tau^{st} + K_0^{st} + \frac{a\Delta t}{\Delta x} K_\xi^{st} \right)^{-1} M_0 \hat{u}_i^n, \tag{14}$$

where the three $N_{st} \times N_{st}$ matrices $K_\tau^{st}, K_0^{st}, K_\xi^{st}$ and the $N_{st} \times (N + 1)$ matrix M_0 are defined as

$$\begin{aligned} [K_\tau^{st}]_{k,\ell} &= \int_0^1 \int_0^1 \theta_k(x,t) \partial_t \theta_\ell(x,t)\, \mathrm{d}x\mathrm{d}t, & [K_\xi^{st}]_{k,\ell} &= \int_0^1 \int_0^1 \theta_k(x,t) \partial_x \theta_\ell(x,t)\, \mathrm{d}x\mathrm{d}t, \\ [K_0^{st}]_{k,\ell} &= \int_0^1 \theta_k(x,0)\theta_\ell(x,0)\, \mathrm{d}x, & [M_0]_{k,\ell} &= \int_0^1 \theta_k(x,0)\phi_\ell(x)\, \mathrm{d}x. \end{aligned} \tag{15}$$

By taking now into account that the Rusanov-type flux (12) for the LAE is a simple upwind scheme and recalling $a > 0$, the subsequent corrector step in (11) simplifies to

$$\hat{u}_i^{n+1} = \hat{u}_i^n + \frac{a\Delta t}{\Delta x} M^{-1} (K_\xi \hat{q}_i^n - F_-^r \hat{q}_i^n + F_-^l \hat{q}_{i-1}^n), \tag{16}$$

where the three $(N + 1) \times N_{st}$ matrices K_ξ, F_-^r, F_-^l and the $(N + 1) \times (N + 1)$ mass matrix M are defined as

$$\begin{aligned} [M]_{k,\ell} &= \int_0^1 \phi_k(x)\phi_\ell(x)\, \mathrm{d}x, & [K_\xi]_{k,\ell} &= \int_0^1 \int_0^1 \partial_x \psi_k(x,t)\theta_\ell(x,t)\, \mathrm{d}x\mathrm{d}t, \\ [F_-^r]_{k,\ell} &= \int_0^1 \psi_k(1,t)\theta_\ell(1,t)\, \mathrm{d}t, & [F_-^l]_{k,\ell} &= \int_0^1 \psi_k(0,t)\theta_\ell(1,t)\, \mathrm{d}t. \end{aligned} \tag{17}$$

Since $\hat{u}_{i-1}^n = e^{-\mathrm{i}\kappa\Delta x}\hat{u}_i^n$, we have from (14) that $\hat{q}_{i-1}^n = e^{-\mathrm{i}\kappa\Delta x}\hat{q}_i^n$. Hence, the update in (16) reduces to

$$\begin{aligned} \hat{u}_i^{n+1} &= \hat{u}_i^n + \frac{a\Delta t}{\Delta x} M^{-1} \left(K_\xi - F_-^r + e^{-\mathrm{i}\kappa\Delta x} F_-^l \right) \hat{q}_i^n \\ &= \hat{u}_i^n + \frac{a\Delta t}{\Delta x} M^{-1} \left(K_\xi - F_-^r + e^{-\mathrm{i}\kappa\Delta x} F_-^l \right) \left(K_\tau^{st} + K_0^{st} + \frac{a\Delta t}{\Delta x} K_\xi^{st} \right)^{-1} M_0 \hat{u}_i^n \\ &= A_N(\mathrm{CFL}, \theta) \cdot \hat{u}_i^n, \end{aligned}$$

Table 1: Increment indicator $\Delta_N^{m\to m+1}$ for varying degrees N and for $m = 1, \dots, 11$. At each step, we evaluate how the discrete amplification factor is affected upon doubling the subdivision of the interval $[0, \pi]$. The low increments motivate the sampling decision $m = 6$.

	$N=1$	$N=2$	$N=3$	$N=4$	$N=5$	$N=6$	$N=7$	$N=8$	$N=9$
$\Delta^{1\to2}$	0	0	0	4.11e-06	9.58e-07	3.54e-05	1.13e-05	8.13e-06	6.72e-09
$\Delta^{2\to3}$	0	0	0	3.02e-08	9.91e-07	3.73e-06	7.02e-06	9.55e-06	6.79e-14
$\Delta^{3\to4}$	0	0	0	3.34e-08	6.28e-07	1.82e-06	2.96e-06	1.46e-06	1.22e-13
$\Delta^{4\to5}$	0	0	0	3.29e-08	2.70e-07	6.53e-07	9.05e-07	7.10e-07	3.51e-13
$\Delta^{5\to6}$	0	0	0	4.33e-08	1.46e-07	1.98e-07	2.05e-07	1.59e-07	2.29e-12
$\Delta^{6\to7}$	0	0	4.44e-16	7.84e-09	3.50e-08	5.52e-08	5.37e-08	3.72e-08	1.55e-11
$\Delta^{7\to8}$	0	0	4.44e-16	2.30e-09	9.38e-09	1.31e-08	1.47e-08	1.07e-08	2.86e-12
$\Delta^{8\to9}$	0	0	4.44e-16	7.10e-10	2.41e-09	3.22e-09	3.42e-09	2.27e-09	7.95e-13
$\Delta^{9\to10}$	0	0	6.66e-16	1.80e-10	6.10e-10	8.92e-10	8.59e-10	6.34e-10	3.28e-13
$\Delta^{10\to11}$	0	0	4.44e-16	4.09e-11	1.48e-10	2.07e-10	2.33e-10	1.67e-10	8.93e-14
$\Delta^{11\to12}$	0	0	4.44e-16	1.13e-11	3.42e-11	5.42e-11	5.26e-11	3.87e-11	8.99e-14

with an amplification matrix $A_N(\mathrm{CFL}, \theta) \in \mathbb{C}^{(N+1)\times(N+1)}$ dependent only on $\theta = \kappa\Delta x$ and $\mathrm{CFL} = a\Delta t/\Delta x$. We observe that every matrix in (15) and (17) can be computed analytically; consequently, the same holds for each amplification matrix A_N. For a matrix A, let us denote by $\rho(A)$ its spectral radius. Then, for a given CFL number, a necessary condition for the update formula for $\hat{u}_i^n$ to be stable, regardless of the value of Δx, is

$$\rho(A_N(\mathrm{CFL}, \theta)) \leq 1 \quad \text{for every } \theta \in [0, \pi]. \tag{18}$$

In (18), the phase angle can be restricted to $[0, \pi]$ since, by construction, $\rho(A_N(\mathrm{CFL}, \theta)) = \rho(A_N(\mathrm{CFL}, -\theta))$ for every $\mathrm{CFL} \geq 0$ and $\theta \in [0, \pi]$.

Numerical setting for studying stability conditions.
As we know that a finite volume scheme requires $\mathrm{CFL} \leq 1$, we consider a uniform subdivision of the CFL domain $[0, 1]$ into 10^4 elements, i.e., we check condition (18) over the discrete sample set $\{c = k10^{-4} \mid k \in \mathbb{N}\} \cap [0, 1]$. Without loss of generality, we can focus our study on the set of CFL values for which the amplification factor related to constant states is no bigger than 1.1. Hence, we define the set

$$C_N := \left\{c = k \cdot 10^{-4} \mid k \in \mathbb{N},\, \rho(A_N(c, 0)) \leq 1.1\right\} \quad \text{for each } N = 1, \dots, 9. \tag{19}$$

To numerically evaluate (18), we consider a finite set of phase angles θ. We subdivide the interval $[0, \pi]$ into 2^m uniform subintervals, for a fixed $m \in \mathbb{N}$, and check condition (18) at the corresponding $2^m + 1$ nodes, obtaining the discrete amplification factor

$$\rho_{N,m}(c) = \max_{k=1,\dots,2^m+1} \rho\left(A_N\left(c, \frac{k-1}{2^m}\pi\right)\right) \quad \text{for } c \in [0, 1]. \tag{20}$$

Since by construction $\rho_{N,m} \leq \rho_{N,m+1}$, to study the effect of the choice of m on the discrete amplification factor, we can define the non-negative *increment indicator*

$$\Delta_N^{m\to m+1} = \max_{c\in C_N} \left\{\rho_{N,m+1}(c) - \rho_{N,m}(c)\right\}, \tag{21}$$

that measures the variation of the discrete maximum when doubling the evaluation points. A thorough study of this increment indicator is reported in Table 1. We can observe how the discrete amplification factor stabilises very quickly as we increase m, independently of the degree N. For $N = 1, 2, 3$, the maximum seems to be achieved at 0 and/or π (as the increment indicator is always zero), while for $N \geq 4$ the discrete amplification factor is not increasing more than 10^{-7} already for $m > 6$. In particular, if we take into account that the relevant amplification factors in Figure 2 are around $1 + 10^{-4}$, in the sequel, unless differently specified, we will always compute discrete maxima by subdividing $[0, \pi]$ into 2^6 uniform elements, i.e., we fix $m = 6$.

Table 2: Relevant CFL bounds for varying polynomial degrees $N = 1, \dots, 9$. First line: commonly used empirical CFL bounds, as reported in [32]. Second line: largest tested CFL value such that $\rho_{N,6}(\text{CFL}) \leq 1 + 10^{-3}$. Third line: largest tested CFL value such that $\rho_{N,11}(\text{CFL}) \leq 1 + 10^{-10}$.

	$N = 1$	$N = 2$	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$	$N = 8$	$N = 9$
$\text{CFL}_{empirical}$	0.33	0.17	0.10	0.069	0.045	0.038	0.03	0.02	0.015
$\text{CFL}_{\rho \leq 1+1e-3}$	0.3334	0.1709	0.1039	0.0698	0.0501	0.0377	0.0293	0.0235	0.0192
$\text{CFL}_{\rho \leq 1+1e-10}$	0.3333	0.1708	0.1039	9.23e-4	5.07e-4	7.76e-4	3.67e-3	9.89e-3	1.58e-3

Identification of relevant CFL bounds.
For each $N = 1, \dots, 9$, we compute the discrete amplification factor $\rho_{N,6}$. We observe that the explicit ADER-DG scheme always preserves constant states: indeed, for $\theta = 0$ and $\hat{u}_i^n = (1, 0, \dots, 0)$, we have $A_N(c, 0) \cdot \hat{u}_i^n = \hat{u}_i^n$, independently of $c \in [0, 1]$. Hence, $\rho_{N,m}(c) \geq 1$ for every c and every m, in particular $\rho_{N,6} \geq 1$. In view of this, we report in Figure 2 the graph of $\rho_{N,6} - 1$. Furthermore, the second row of Table 2 identifies the highest $c \in C_N$ such that $\rho_{N,6}(c) \leq 1 + 10^{-3}$. For comparison, the first row recalls the CFL bounds widely used in the community (see, e.g., [32, 13, 67]), which have generally been determined empirically.

For $N \leq 3$, the empirical CFL bounds and the CFL bounds determined by our linear stability analysis coincide, in accordance with the analysis carried out in [16]. On the other hand, for $4 \leq N \leq 9$, the method turns out to be unstable for commonly used CFL bounds. By comparing the first and second rows in Table 2, we observe that empirical CFL bounds just correspond to the rightmost jump in each stability plot, where the stability condition (18), even if only slightly, is violated. Hence, commonly used CFL values do not correspond to true stability limits. However, this does not mean that they cannot be used in applications as an informed guess for the selection of the time step. Indeed, in complex test cases, the small amplification factor due to instability is mitigated when combined with other discretization techniques such as numerical viscosity, Riemann solvers, damping factors, and limiters. As a result, authors usually adopt these values as their reference maximum CFL number when running numerical experiments.

A sharp von Neumann limit can be recovered for $N \geq 4$ only for much smaller values of the CFL number, as shown in Figure 3, where we pick a uniform CFL sampling in the log-domain (starting from 10^{-12} with a log-step of 0.005). We report in the third and last row of Table 2 the largest tested CFL value c such that $\rho_{N,11}(c) \leq 1 + 10^{-10}$.

Remark 1. The analysis extends to the case of a mesh moving with uniform velocity $v \in \mathbb{R}$. In this context, the stability condition turns into

$$|a - v| \frac{\Delta t}{\Delta x} \leq \text{CFL}_{\max}.$$

Assuming $a > 0$, by using a value $\tilde{c} \leq \text{CFL}_{\max}$ and selecting $\Delta t = \tilde{c} \Delta x / a$, the condition above rewrites as

$$a \left(1 - \frac{\text{CFL}_{\max}}{\tilde{c}}\right) \leq v \leq a \left(1 + \frac{\text{CFL}_{\max}}{\tilde{c}}\right),$$

which provides a bound on the admissible velocities of the grid. For example, if $\tilde{c} = \text{CFL}_{\max}$ one obtains $0 \leq v \leq 2a$, while if $\tilde{c} = 0.5 \cdot \text{CFL}_{\max}$ the condition becomes $-a \leq v \leq 3a$, allowing more freedom in the mesh velocities. This suggests the time step selection rule

$$\Delta t^n \leq \tilde{c} \min_{i=1,\dots,N_e} \frac{\Delta x_i^n}{\max\{|f'(u_i^n(x_{i-1/2}^n))|, |f'(u_i^n(x_{i+1/2}^n))|\}},$$

without explicitly incorporating the mesh velocity, but accounting for it implicitly by selecting a suitable $\tilde{c}$.

Remark 2. Recently, [63] independently derived the CFL bounds in the second row of Table 2 through a thorough analytical treatment of the amplification matrix A_N, under a suitable ansatz on the behaviour of its eigenvalues. Despite employing spacetime tensor basis functions, their von Neumann stability analysis retrieves identical bounds and yields the same conclusions regarding the stability of the explicit ADER-DG scheme.

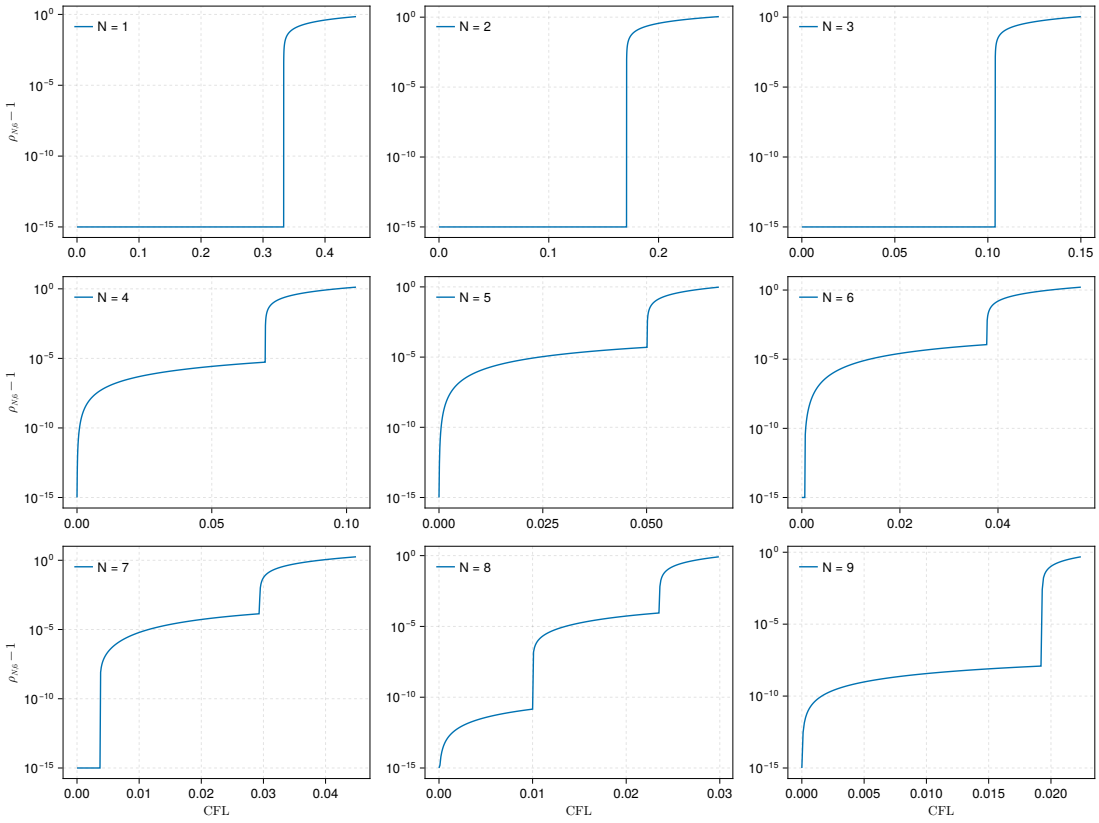


Figure 2: Semi-logarithmic plot of the function CFL $\mapsto \rho_{N,6}(\text{CFL}) - 1$, where the discrete maximal spectral radius is computed over $2^6 + 1$ equispaced points over $[0, \pi]$. The necessary stability condition for the explicit ADER-DG method requires this function to vanish, see (18).

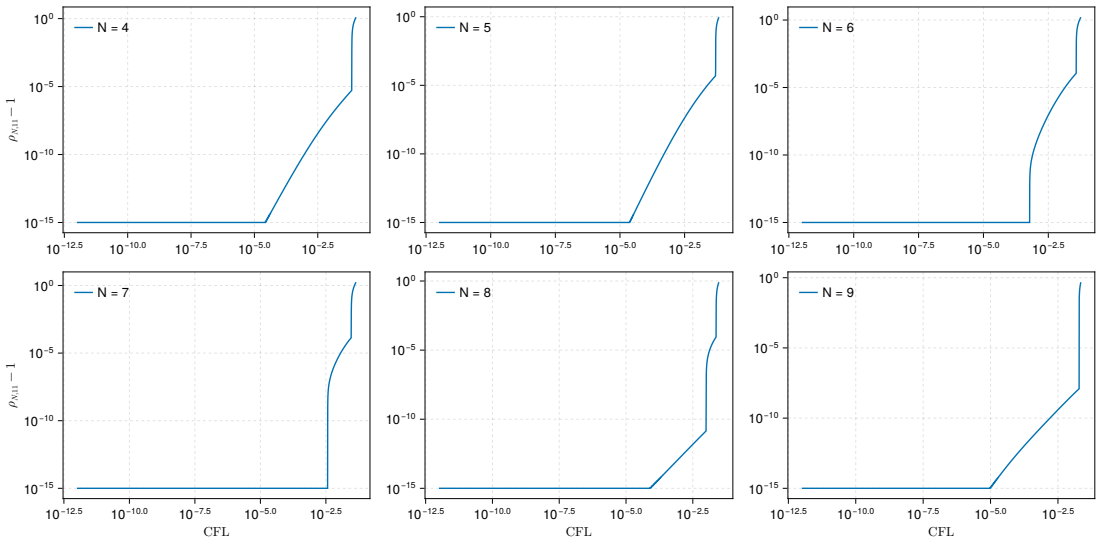


Figure 3: Logarithmic plot of the function CFL $\mapsto \rho_{N,11}(\text{CFL}) - 1$, where the maximum spectral radius is computed over $2^{11} + 1$ equispaced points over $[0, \pi]$. Two clear corner points identify the stability threshold that often corresponds to very low CFL values, while the larger one is the typically used empirical CFL bound.

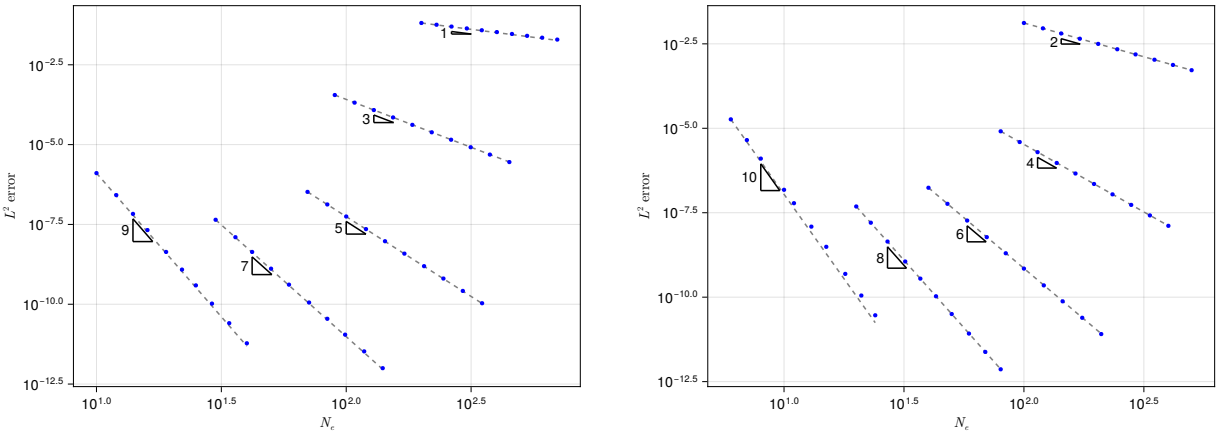


Figure 4: Convergence order of the explicit ADER-DG method: left odd orders for $N = 0, 2, 4, 6, 8$, right even orders for $N = 1, 3, 5, 7, 9$.

3.2. Numerical convergence order of the explicit method

The explicit ADER-DG method is expected to have order of convergence $N + 1$ for a given polynomial degree N. We verify this property by evolving a linear advection equation with unit velocity on the domain $[-6, 6]$, with periodic boundary conditions and starting from $Q(x, 0) = e^{-x^2}$. We vary the number of elements N_e and we consider a CFL at $0.9 \cdot \text{CFL}_{\max}$, where $\text{CFL}_{\max}$ is taken from the second row of Table 2. In Figure 4, we report, in logarithmic scale, the L^2 norm of the error with respect to the exact solution at time $T = 12$, i.e., after a full loop. Each convergence order is correctly achieved as expected.

3.3. Stability of the implicit method

The stability analysis for the implicit ADER-DG method can be developed following the same steps used for the explicit counterpart. Again, we have to study the spectral radius of a suitable amplification matrix $A_N(\text{CFL}, \theta)$, which can be computed exactly. We evaluate condition (18) by means of computing $\rho_{N,6}$, see (20), on the discrete set $C = \{c = k \cdot 10^{-2} \mid k \in \mathbb{N}\} \cap [0, 10]$. We obtain the following:

	$N = 1$	$N = 2$	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$	$N = 8$	$N = 9$
$\max_{c \in C} \rho_{N,6}(c) - 1$	0	8.88e-16	2.89e-15	3.55e-15	4.00e-15	2.95e-13	2.38e-14	2.52e-13	2.03e-12

No clear CFL-type bounds can be identified, suggesting unconditional stability. This numerical check is coherent with the L^2 stability of the scheme provided in [17]. In the specific von Neumann setting, we can actually prove stability of the scheme for a slightly different choice of spacetime basis functions, namely tensor basis functions. This is possible because, in this new framework, the method can be cast as a Runge–Kutta method whose stability was analysed in [57]. While the choice of spacetime basis functions is different, the scheme is expected to behave similarly (see also the recent results in [63], where explicit stability bounds for the explicit ADER-DG method were obtained in a spacetime tensor basis framework). We can prove the following.

Theorem 1. *Consider the linear advection equation*

$$\partial_t u(x, t) + \partial_x u(x, t) = 0 \quad \text{for } x \in (0, 2\pi), t > 0,$$

coupled with periodic boundary conditions and initial datum $u(x, 0) = e^{\mathrm{i}\kappa x}$, $\kappa \in \mathbb{N}$. *Consider over* $[0, 2\pi]$ *a uniform Eulerian grid of* $N_e > 0$ *elements of size* $\Delta x > 0$. *Consider the Rusanov-type numerical flux* (12). *Then, when applied to this problem, the implicit ADER-DG method as described in* (13)*, with spacetime tensor basis functions, is stable for any* $\Delta t > 0$.

Proof. In the uniform Eulerian case, we have that $\Omega_i^{n+1} = \Omega_i^n = \Omega_i$, each spatial element has measure $\Delta x > 0$ and $C_i^n = \Omega_i \times [t^n, t^{n+1}]$. In space, over each element Ω_i we can use Lagrange basis functions $\{\psi_{i,k_s}^s\}_{k_s=0}^N$ defined on Gauss–Legendre points, and similarly in time we use Lagrange basis functions $\{\psi_{n,k_t}^t\}_{k_t=0}^N$ defined on Gauss–Legendre points over $[t^n, t^{n+1}]$. Indeed, this choice is coherent with the one made in Section 2 because it has been shown that any polynomial choice with exact quadrature leads to an equivalent method [36]. Spacetime basis functions are then defined in a tensor fashion as $\theta_{i,(k_s,k_t)}^n(x,t) = \psi_{i,k_s}^s(x)\psi_{n,k_t}^t(t)$, for indices $k_s, k_t \in \{0,\dots,N\}$. In particular, approximants take now the form

$$q_i^n(x,t) = \sum_{k_s,k_t=0}^{N} \theta_{i,(k_s,k_t)}^n(x,t)\hat{q}_{i,(k_s,k_t)}^n \quad \text{and} \quad u_i^n(x) = \sum_{k_s=0}^{N} \psi_{i,k_s}^s(x)\hat{u}_{i,k_s}^n$$

for given coefficients $\{\hat{q}_{i,(k_s,k_t)}^n\}_{k_s,k_t} \subset \mathbb{R}$ and $\{\hat{u}_{i,k_s}^n\}_{k_s} \subset \mathbb{R}$. In this setting, the implicit ADER-DG method in (13) can be written, using integration by parts in time, as

$$\int_{\Omega_i} \theta_{i,(k_s,k_t)}^n(x,t^{n+1})q_i^n(x,t^{n+1})\,\mathrm{d}x - \int_{\Omega_i} \theta_{i,(k_s,k_t)}^n(x,t^n)u_i^n(x)\,\mathrm{d}x - \int_{C_i^n} \partial_t\theta_{i,(k_s,k_t)}^n(x,t)q_i^n(x,t)\,\mathrm{d}x\mathrm{d}t$$
$$- \int_{C_i^n} \partial_x\theta_{i,(k_s,k_t)}^n(x,t)q_i^n(x,t)\,\mathrm{d}x\mathrm{d}t + \int_{t^n}^{t^{n+1}} \left(\theta_{i,(k_s,k_t)}^n(x_{i+\frac{1}{2}},t)q_i^n(x_{i+\frac{1}{2}},t) - \theta_{i,(k_s,k_t)}^n(x_{i-\frac{1}{2}},t)q_{i-1}^n(x_{i-\frac{1}{2}},t)\right)\mathrm{d}t = 0$$

for $i = 1,\dots,N_e$ and $k_s, k_t = 0,\dots,N$. Now, by using the von Neumann ansatz that $q_{i-1}^n = e^{-\mathrm{i}\theta}q_i^n$, $\theta = \kappa\Delta x$, and omitting the cell indexes i, n, we define

$$[M^s]_{k_s,j_s} := \int_{\Omega_i} \psi_{k_s}^s\psi_{j_s}^s\,\mathrm{d}x, \qquad [M^t]_{k_t,j_t} := \int_{t^n}^{t^{n+1}} \psi_{k_t}^t\psi_{j_t}^t\,\mathrm{d}t, \qquad [E^t]_{k_t} := \psi_{k_t}^t(t^n),$$
$$[D^s]_{k_s,j_s} := \psi_{k_s}^s(x_{i+\frac{1}{2}})\psi_{j_s}^s(x_{i+\frac{1}{2}}) - \psi_{k_s}^s(x_{i-\frac{1}{2}})e^{-\mathrm{i}\theta}\psi_{j_s}^s(x_{i+\frac{1}{2}}) - \int_{\Omega_i} \partial_x\psi_{k_s}^s(x)\psi_{j_s}^s(x)\,\mathrm{d}x,$$
$$[A^t]_{k_t,j_t} := \psi_{k_t}^t(t^{n+1})\psi_{j_t}^t(t^{n+1}) - \int_{t^n}^{t^{n+1}} \partial_t\psi_{k_t}^t(t)\psi_{j_t}^t(t)\,\mathrm{d}t,$$

so that the scheme becomes, with Einstein notation,

$$[A^t]_{k_t,j_t}[M^s]_{k_s,j_s}\hat{q}_{(j_s,j_t)} - [M^s]_{k_s,j_s}[E^t]_{k_t}\hat{u}_{j_s} + [M^t]_{k_t,j_t}[D^s]_{k_s,j_s}\hat{q}_{(j_s,j_t)} = 0.$$

By inverting the mass matrices, and using the fact that $[A^t]^{-1}E^t$ gives a vector of ones [36], we obtain

$$\hat{q}_{(z_s,z_t)} - \hat{u}_{z_s} + ([A^t]_{z_t,k_t}^{-1}[M^t]_{k_t,j_t})([M^s]_{z_s,k_s}^{-1}[D^s]_{k_s,j_s})\hat{q}_{(j_s,j_t)} = 0.$$

This can be written as a Runge–Kutta scheme with time stages k_t for the ODE semidiscrete problem

$$\partial_t\hat{q}_{z_s} + ([M^s]_{z_s,k_s}^{-1}[D^s]_{k_s,j_s})\hat{q}_{j_s} = 0.$$

In particular, it has been shown [57] that the Runge–Kutta method given by the matrix $[A^t]_{z_t,k_t}^{-1}[M^t]_{k_t,j_t}$ is A-stable. So, the method is unconditionally stable if the real part of the eigenvalues of $B = -[M^s]^{-1}\cdot D^s \in \mathbb{C}^{(N+1)\times(N+1)}$ are all nonpositive. We prove this condition by showing that B is negative semidefinite in the scalar product defined by $\langle\hat{v},\hat{w}\rangle_{M^s} := \hat{v}^*M^s\hat{w}$ because for Lagrange basis functions the spatial mass matrix is positive definite (here, $\hat{v}^*$ denotes the conjugate transpose). So, for any $\hat{v} \in \mathbb{C}^{N+1}$, upon defining $v(x) := \hat{v}_{j_s}\psi_{j_s}^s(x)$, we have that

$$\begin{aligned}
\mathrm{Re}\left(\left\langle\hat{v}, -[M^s]^{-1}[D^s]\hat{v}\right\rangle_{M^s}\right) &= \mathrm{Re}\,(-\hat{v}^*D^s\hat{v}) = \mathrm{Re}\left(-\overline{v(x_{i+1/2})}v(x_{i+1/2}) + \overline{v(x_{i-1/2})}e^{-\mathrm{i}\theta}v(x_{i+1/2}) + \int_{\Omega_i}\partial_x\overline{v(x)}\cdot v(x)\,\mathrm{d}x\right)\\
&= -|v(x_{i+1/2})|^2 + \mathrm{Re}\left(\overline{v(x_{i-1/2})}e^{-\mathrm{i}\theta}v(x_{i+1/2})\right) + \mathrm{Re}\left(\int_{\Omega_i}\partial_x\overline{v(x)}\cdot v(x)\,\mathrm{d}x\right)\\
&= -|v(x_{i+1/2})|^2 + \mathrm{Re}\left(\overline{v(x_{i-1/2})}e^{-\mathrm{i}\theta}v(x_{i+1/2})\right) + \frac{|v(x_{i+1/2})|^2}{2} - \frac{|v(x_{i-1/2})|^2}{2}\\
&\le -\frac{|v(x_{i+1/2})|^2}{2} - \frac{|v(x_{i-1/2})|^2}{2} + |v(x_{i-1/2})|\,|v(x_{i+1/2})| = -\frac{1}{2}(|v(x_{i-1/2})| - |v(x_{i+1/2})|)^2 \le 0.
\end{aligned}$$

Table 3: We consider a unit-velocity LAE over $[-6,6]$ with initial condition e^{-x^2} at time $t = 0$. The spatial discretization is fixed at $N_e = 24$ elements, and the implicit method is run for varying degrees N and varying CFL values, leading to a time step $\Delta t = \Delta x \cdot \text{CFL} = 0.5 \cdot \text{CFL}$. We report the L^2-error of the numerical solution at final time $T = 12$, after $N_t = T/\Delta t = 24/\text{CFL}$ time steps are performed. We observe how the method remains stable but progressively becomes more and more dissipative as CFL increases.

	CFL = 0.2	CFL = 0.5	CFL = 1.0	CFL = 2.0	CFL = 3.0	CFL = 6.0	CFL = 12.0
$N = 1$	1.82e-01	2.49e-01	3.49e-01	5.13e-01	6.29e-01	8.07e-01	9.14e-01
$N = 2$	1.05e-02	1.98e-02	3.68e-02	1.76e-01	3.13e-01	5.81e-01	8.40e-01
$N = 3$	4.60e-04	1.12e-03	3.25e-03	3.25e-02	9.48e-02	3.73e-01	6.73e-01
$N = 4$	3.40e-05	9.31e-05	8.17e-04	3.69e-03	2.67e-02	2.01e-01	5.54e-01
$N = 5$	2.83e-06	9.89e-06	1.50e-04	7.66e-04	5.28e-03	9.93e-02	4.28e-01
$N = 6$	2.08e-07	1.20e-06	9.76e-06	2.91e-04	9.47e-04	4.17e-02	3.12e-01
$N = 7$	1.47e-08	1.69e-07	1.38e-06	7.13e-05	2.85e-04	1.68e-02	2.22e-01
$N = 8$	1.05e-09	2.35e-08	2.38e-07	6.86e-06	1.05e-04	5.68e-03	1.50e-01
$N = 9$	6.18e-11	1.39e-09	2.57e-08	7.33e-07	2.46e-05	2.26e-03	9.70e-02

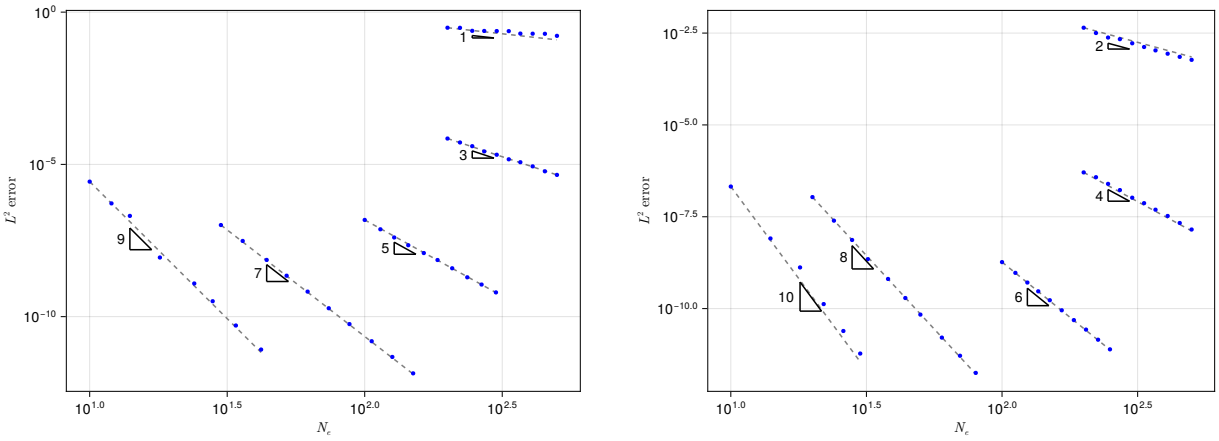


Figure 5: Convergence order of the implicit ADER-DG method: left odd orders for $N = 0, 2, 4, 6, 8$, right even orders for $N = 1, 3, 5, 7, 9$.

This shows that all eigenvalues of the semidiscrete operator fall in the left complex half-plane, hence, the method is von Neumann stable independently of the time step. □

3.4. Numerical convergence order and accuracy of the implicit method

The implicit ADER-DG method, as its explicit counterpart, is expected to have order of convergence $N + 1$ for a given polynomial degree N. We verify this by means of the same LAE evolution test case described in Section 3.2. We vary the number of elements N_e and we consider a CFL at $10 \cdot \text{CFL}_{\text{max}}$ where CFL_{max} is taken from the second row of Table 2. In Figure 5, we report, in logarithmic scale, the L^2 norm of the error with respect to the exact solution at time $T = 1$. Each convergence order is correctly achieved as expected.

While the method is expected to be stable for any CFL number, arbitrarily large values cannot be chosen if an accurate long-term simulation is desired. This is illustrated in Table 3, where again the same LAE evolution test case described in Section 3.2 is evaluated on a fixed mesh of $N_e = 24$ spatial elements up to $T = 12$ (corresponding to one full domain revolution). As the CFL increases, the scheme requires fewer time steps and, while remaining stable, progressively loses accuracy due to increased numerical dissipation.

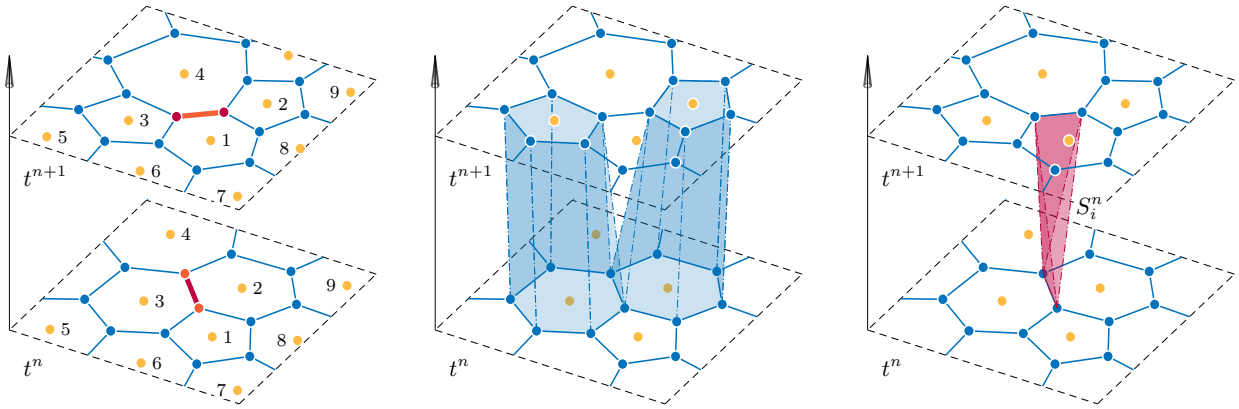


Figure 6: Spacetime connectivity with a topology change in two dimensions (left), induced hole-like sliver element (right) and neighbouring classical control volumes (middle).

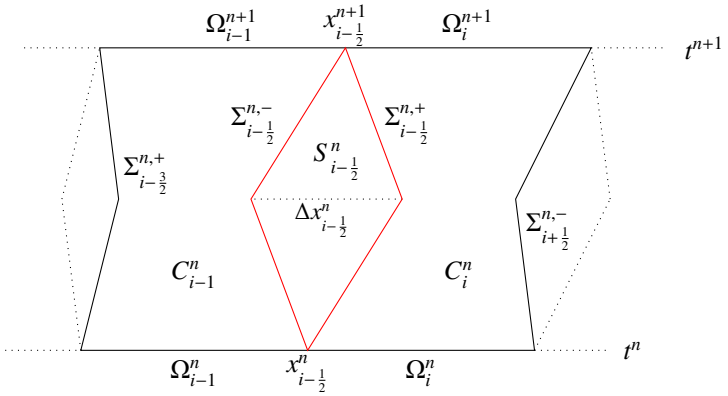


Figure 7: Prototype configuration of a spacetime domain discretization with a degenerate (or sliver) element. At each time t^n the spatial domain is subdivided into elements Ω_i^n, connected in spacetime by control volumes C_i^n. Interfaces between control volumes are expanded into hole-like sliver elements $S_{i-1/2}^n$ with maximal width $\Delta x_{i-1/2}^n$, mimicking the 3d hole-like sliver elements described in Figure 6.

4. ALE ADER-DG methods on degenerate spacetime geometries

In this section, we consider a modification of the ALE ADER-DG scheme on degenerate spacetime geometries, which are obtained by introducing in our computational domain *fictitious spacetime degenerate elements*, which will be termed *sliver* elements.

Geometry description.
To construct a *degenerate* spacetime geometry we start from the classical geometry described in Figure 1. As in Figure 7, we assume now that some of the interfaces $\Sigma_{i-1/2}^n$ are expanded into a *sliver element* $S_{i-1/2}^n$, which represents our fictitious spacetime degenerate element. To construct a sliver element $S_{i-1/2}^n$, we first fix its maximal width $\Delta x_{i-1/2}^n > 0$ and then consider the quadrilateral whose vertices, in counter-clockwise order, are defined as

$$(x_{i-1/2}^n, t^n), \quad ((x_{i-1/2}^n + x_{i-1/2}^{n+1} + \Delta x_{i-1/2}^n)/2, t^n + \Delta t^n/2),$$
$$(x_{i-1/2}^{n+1}, t^{n+1}), \quad ((x_{i-1/2}^n + x_{i-1/2}^{n+1} - \Delta x_{i-1/2}^n)/2, t^n + \Delta t^n/2).$$

Each neighbouring control volume is redefined accordingly. The boundary $\partial S_{i-1/2}^n$ of each sliver element can be viewed as the union of $\Sigma_{i-1/2}^{n,-}$ (the interface between $S_{i-1/2}^n$ and C_{i-1}^n) and $\Sigma_{i-1/2}^{n,+}$ (the interface between $S_{i-1/2}^n$ and C_i^n). In case an interface $\Sigma_{i-1/2}^n$ is not replaced by a sliver element, then we set $\Sigma_{i-1/2}^{n,\pm} := \Sigma_{i-1/2}^n$.

Remark 3. These degenerate elements mimic in 1d the hole-like sliver elements introduced in [30, 31, 28, 29, 33] within the framework of 2d/3d moving meshes with topology changes, see Figure 6. Those works are all based on the use of a direct ALE approach for the evolution of the PDE and therefore on the integration over control volumes that connect in time two different meshes. When, in this context, the meshes do not merely move but can also change topology, meaning that connectivity and list of neighbours of each cell may change, the use of classical control volumes to connect the meshes is no longer sufficient. To cover the entire spacetime domain in between, it then becomes necessary to introduce additional volumes, such as those shown in the right panel of Figure 6, which, similarly to the 1d degenerate elements of this paper, have zero area at both times t^n and t^{n+1}, but a non zero spacetime volume. At present, the 2d sliver elements have been used in a variety of applications, but a theoretical analysis of their properties has not yet been carried out in detail.

Basis functions.
For each $n > 0$ and $i \in \{1, \dots, N_e\}$, we consider the already introduced families of basis functions $\{\phi^n_{i,\ell}\}_\ell$, $\{\theta^n_{i,\ell}\}_\ell$ and $\{\psi^n_{i,\ell}\}_\ell$, see (2), (4) and (3). Furthermore, we also consider over each sliver element $S^n_{i-1/2}$ the family of *spacetime* basis function $\{\theta^n_{i-1/2,\ell}\}_\ell$ defined as

$$\theta^n_{i-\frac{1}{2},\ell} \colon S^n_{i-\frac{1}{2}} \to \mathbb{R},\ (x,t) \mapsto \left(\frac{x - x^n_{i-\frac{1}{2}}}{\Delta x^n_{i-\frac{1}{2}}}\right)^{\ell_1} \left(\frac{t - t^n}{\Delta t^n}\right)^{\ell_2}$$
$$\text{for } \ell = \ell_1 + \ell_2(2N + 3 - \ell_2)/2,\ 0 \le \ell_1 + \ell_2 \le N.$$

These are nothing else than modal spacetime basis functions centred at $(x^n_{i-1/2}, t^n)$ and scaled according to $\Delta x^n_{i-1/2}$ and Δt^n. Alongside the general approximants u^n_i and q^n_i described in (5) and (6), we also consider approximants defined on each sliver element $S^n_{i-1/2}$ to take the form of a spacetime polynomial function $q^n_{i-1/2}$ written as

$$q^n_{i-\frac{1}{2}}(x,t) = \sum_{\ell=0}^{N_{st}-1} \theta^n_{i-\frac{1}{2},\ell}(x,t)\hat{q}^n_{i-\frac{1}{2},\ell} \quad \text{for } (x,t) \in S^n_{i-\frac{1}{2}},\ i \in \{2, \dots, N_e\}, n > 0, \tag{22}$$

where $\{\hat{q}^n_{i-1/2,\ell}\}_{n,i,\ell} \subset \mathbb{R}$ are the degrees of freedom. Again, we write $\hat{q}^n_{i-1/2}$ to identify the corresponding vector of coefficients.

4.1. Explicit ALE ADER-DG on degenerate geometries

The explicit ALE ADER-DG scheme on degenerate geometries can be described as a three steps scheme, where the third additional step is devoted to the treatment of the newly introduced sliver elements.

Predictor step on control volumes.
The predictor step of the explicit ALE ADER-DG scheme on non-degenerate control volumes remains essentially the same as the one described in Section 2.1.1. The only difference is the possible non-quadrilateral concave structure of some control volumes C^n_i that should be accounted for during the numerical integration. Thus, we can easily apply (10) to obtain each predictor q^n_i, $i = 1, \dots, N_e$.

Locally implicit step on sliver elements.
Once predictors on non-degenerate control volumes are computed, we consider the sliver elements. Let us fix a sliver element $S^n_{i-1/2}$ for some $i \in \{2, \dots, N_e\}$. On such an element, since we have no inflow information to work with at time t^n (due to the sliver degeneracy), we perform a complete integration by parts that allows to introduce boundary fluxes through flux exchanges at the interfaces $\Sigma^{n,\pm}_{i-1/2}$. Hence, we follow the same derivation that led to (13) and we seek for an approximant $q^n_{i-1/2}$ of the form (22) that solves

$$\begin{aligned}
&\int_{\Sigma^{n,-}_{i-\frac{1}{2}}} \theta^n_{i-\frac{1}{2},k} \mathcal{F}(q^n_{i-\frac{1}{2}}, q^n_{i-1}, \hat{\mathbf{n}}^n_{i-\frac{1}{2}})\, \mathrm{d}s + \int_{\Sigma^{n,+}_{i-\frac{1}{2}}} \theta^n_{i-\frac{1}{2},k} \mathcal{F}(q^n_{i-\frac{1}{2}}, q^n_i, \hat{\mathbf{n}}^n_{i-\frac{1}{2}})\, \mathrm{d}s \\
&= \int_{S^n_{i-\frac{1}{2}}} \partial_t \theta^n_{i-\frac{1}{2},k}(x,t) q^n_{i-\frac{1}{2}}(x,t)\, \mathrm{d}x\mathrm{d}t + \int_{S^n_{i-\frac{1}{2}}} \partial_x \theta^n_{i-\frac{1}{2},k}(x,t) f^n_{i-\frac{1}{2}}(x,t)\, \mathrm{d}x\mathrm{d}t \quad \text{for } k = 0, \dots, N_{st} - 1,
\end{aligned} \tag{23}$$

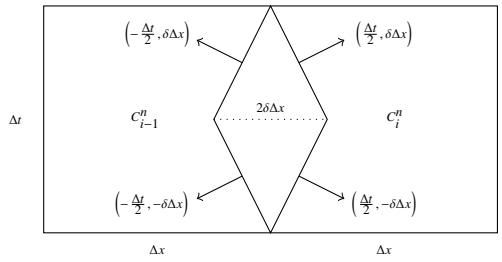


Figure 8: Periodic building block for the von Neumann analysis: we start from a classical Eulerian geometry and replace every other vertical interface with a sliver-like element, whose diameter is controlled by a parameter $\delta \in [0, 0.5]$. Up to scaling, outgoing normal vectors from the sliver-like element take the form reported in the picture.

where $f^n_{i-1/2}$ is defined as in (9) (by replacing the indices accordingly) and $\hat{\mathbf{n}}^n_{i-1/2}$ is the outward pointing normal to $\partial S^n_{i-1/2}$. By taking into account that predictors on the neighbouring non-degenerate control volumes have already been computed, we can approximate the only unknown $\hat{q}^n_{i-1/2}$ by means of solving the algebraic system in (23).

Corrector step.
On each element Ω^{n+1}_i, $i = 1, \dots, N_e$, the new approximant u^{n+1}_i is then computed by means of (11), with the needed substitutions of predictors and interfaces whenever the control volume neighbours a sliver element. We remark that no additional computations are required on sliver elements because they have zero measure boundary at time t^{n+1}. Moreover, the method is conservative by construction also around sliver elements thanks to the exchange of fluxes already performed in the computation of their predictors: indeed, the balance of the fluxes can be obtained by taking $\theta^n_{i-1/2,0} = 1$ as test function in (23).

4.2. Implicit ALE ADER-DG on degenerate geometries

To recover the implicit version of the ALE ADER-DG scheme on degenerate geometries, it is enough to consider together (13) on non-degenerate elements and (23) on sliver elements, by paying attention to use the correct neighbours and interfaces in (13). Again, each new approximant u^{n+1}_i is then recovered by means of an L^2 projection of $q^n_i(\cdot, t^{n+1})$ with respect to the basis $\{\phi^{n+1}_{i,\ell}\}_\ell$.

4.3. Stability analysis with spacetime hole-like sliver elements

The von Neumann stability analysis can be performed in the same fashion as outlined in Section 3. Again, we consider an initial datum of the form $e^{\mathrm{i}\kappa x}$. We build our domain starting from a classical Eulerian geometry and replace every other vertical interface with a sliver-like element, whose diameter is controlled by a parameter $\delta \in [0, 0.5]$ (see Figure 8). Now, we identify two subsequent elements $i-1$ and i separated by a sliver to be our unitary periodic block.

Explicit method.
We study the update of the couple $\hat{v}^n_i = (\hat{u}^n_{i-1}, \hat{u}^n_i)$, which encodes the coefficients of the approximate solution on the given periodic block. While the derivation is somehow cumbersome, we can easily prove that the vector $\hat{v}^n_i \in \mathbb{R}^{2(N+1)}$ follows an update rule of the form

$$\hat{v}^{n+1}_i = B_N(\mathrm{CFL}, \delta, \theta) \cdot \hat{v}^n_i,$$

with an amplification matrix $B_N(\mathrm{CFL}, \delta, \theta) \in \mathbb{C}^{2(N+1)\times 2(N+1)}$ dependent only on $\theta = \kappa\Delta x$, on the parameter δ controlling the width of the sliver and on $\mathrm{CFL} = a\Delta t/\Delta x$. Thus, for a fixed couple (CFL, δ), a necessary condition for the update formula for $\hat{v}^n_i$ to be stable, regardless of the value of Δx, is

$$\rho(B_N(\mathrm{CFL}, \delta, \theta)) \leq 1 \quad \text{for every } \theta \in [0, \pi]. \tag{24}$$

Following the same approach as in Section 3, we introduce, for each $m \in \mathbb{N}$, the discrete amplification factor

$$\rho^\delta_{N,m}(c) = \max_{k=1,\dots,2^m+1} \rho\left(B_N\left(c, \delta, \frac{k-1}{2^m}\pi\right)\right) \quad \text{for } c \in [0, 1], \delta \in [0, 0.5]. \tag{25}$$

We report in Figure 9 the function $\rho^\delta_{N,6} - 1$ for varying $\delta = 0, 0.002, 0.004, 0.006, 0.008$. We observe that the discrete amplification factor for $\delta \approx 0$ always lies below the classical amplification factor $\rho_{N,6} = \rho^0_{N,6}$. Hence, the introduction of the sliver element does not increase the amplification factor of the classical geometry, i.e., the same

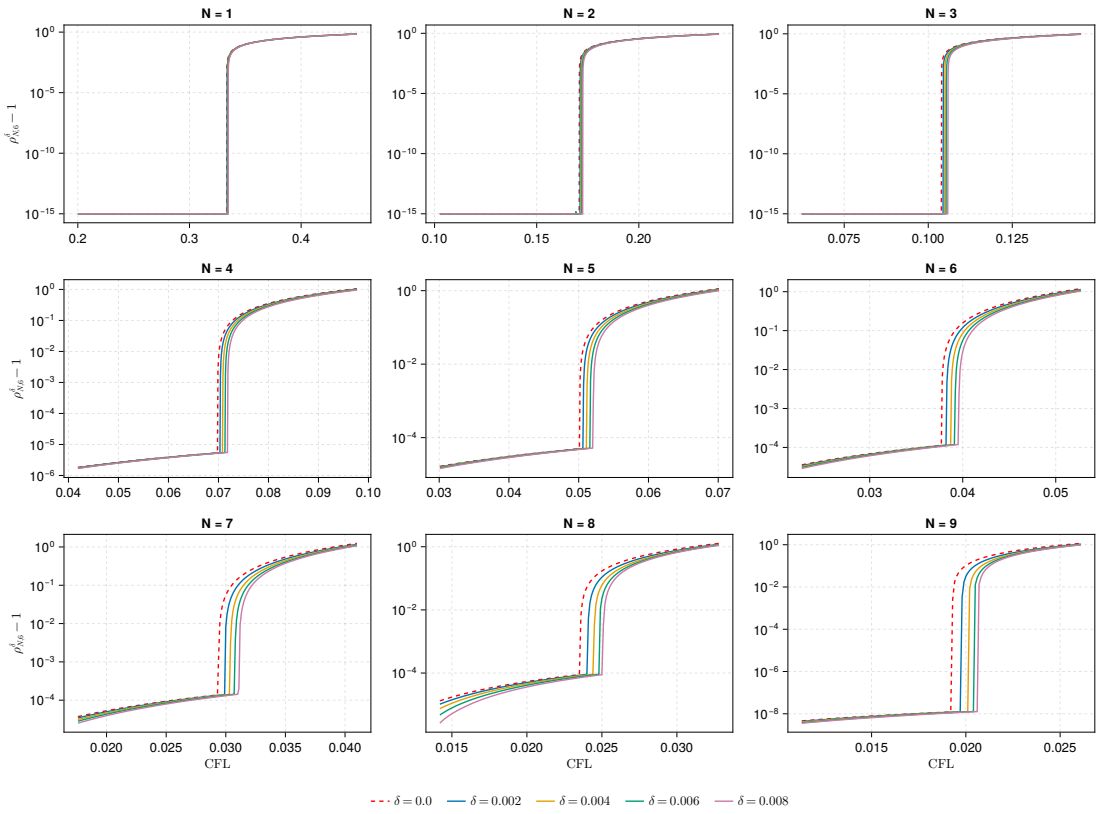


Figure 9: Study of the amplification factor for the explicit ADER-DG method on degenerate geometries. In each plot, we display the function $\text{CFL} \mapsto \rho^0_{N,6}(\text{CFL}) - 1 = \rho_{N,6}(\text{CFL}) - 1$ and the functions $\text{CFL} \mapsto \rho^\delta_{N,6}(\text{CFL}) - 1$ for $\delta = 0.002, 0.004, 0.006, 0.008$. For each CFL the amplification factor in the degenerate setting is always smaller than the one in the classical setting.

CFL bounds that we deem acceptable in the classical setting can be extended to the degenerate setting. We report the full analysis in Figure 10, where we identify the boundary at which $\rho^\delta_{N,6}$ crosses the level $1 + 10^{-3}$ (according to the strategy employed to determine the second row in Table 2). We observe how the introduction of the hole-like sliver elements is slightly increasing the maximal acceptable CFL value: this result is probably a consequence of the implicit treatment of the hole-like element. Indeed, the predictor step on the hole-like sliver element is purely implicit, introducing a locally implicit step in our globally explicit scheme.

Implicit method.
The same von Neumann stability analysis can be developed for the implicit ADER-DG method, by focusing again on the update formula for the couple $\hat{v}^n_i = (\hat{u}^n_{i-1}, \hat{u}^n_i)$ and obtaining the corresponding amplification matrix $B_N(\text{CFL}, \delta, \theta)$. We evaluate condition (24) by means of computing $\rho^\delta_{N,6}$, see (25), on the discrete sets $C = \{c = k \cdot 10^{-2} \mid k \in \mathbb{N}\} \cap [0, 10]$ and $\Delta = \{\delta = k \cdot 10^{-2} \mid k \in \mathbb{N}\} \cap [0, 0.5]$. We obtain the following:

	$N = 1$	$N = 2$	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$	$N = 8$	$N = 9$
$\max_{c \in C, \delta \in \Delta} \rho^\delta_{N,6}(c) - 1$	3.99e-15	5.55e-15	7.55e-15	2.15e-14	1.27e-13	2.01e-12	1.35e-10	4.44e-9	1.21e-7

As is the case in the classical setting, no clear CFL-type bounds can be identified. The largest discrepancy for increasing N can be related to the growing dimension of the matrices involved in the computations, which scale as $3N_{st} \times 3N_{st}$ and reach 165×165 for $N = 9$. These results suggest the unconditional stability of the method. Indeed, following the same idea outlined in the stability proofs in [17, 76, 77], we can actually prove that the method is L^2 stable in several configurations of interest.

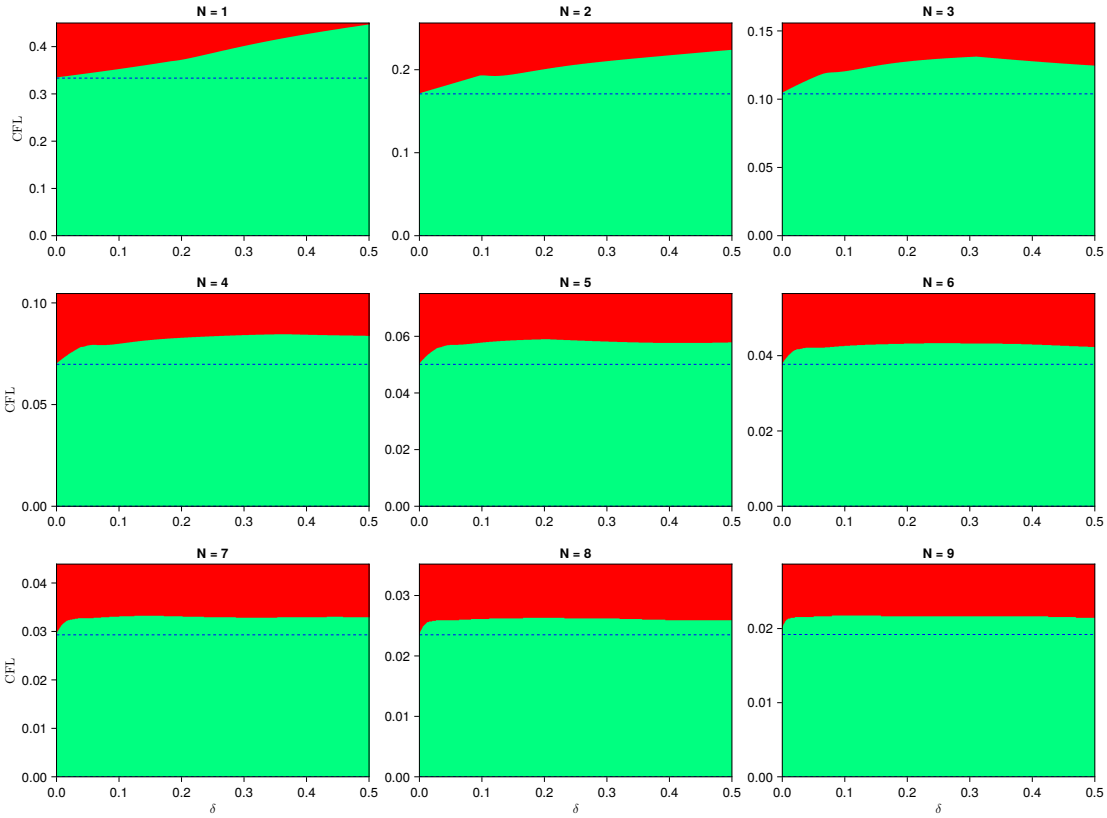


Figure 10: Stability study of the explicit ADER-DG method with slivers. The blue dotted lines are set at the $\text{CFL}_{\rho\leq 1+1e-3}$ values reported in the second row of Table 2. Then, in each plot, a point (δ, CFL) is either green or red depending on whether $\rho^{\delta}_{N,6}(\text{CFL}) \leq 1 + 10^{-3}$ or not. We deduce that classical CFL numbers can be used even in presence of sliver elements.

Theorem 2. *Consider the linear advection equation*

$$\partial_t u(x,t) + \partial_x u(x,t) = 0 \quad \textit{for } x \in (0, 2\pi), t > 0,$$

coupled with periodic boundary conditions and a periodic real valued initial datum $u(x,0) = u_0(x)$*. Consider over* $[0, 2\pi]$ *a uniform Eulerian grid of an even number* $N_e > 0$ *of elements of size* $\Delta x > 0$*. Replace every other interface with a hole-like sliver element of size* $2\delta\Delta x$*,* $\delta \in (0,1)$ *(see Figure 8). Consider the Rusanov-type numerical flux* (12)*. Then, when applied to this problem, the implicit ADER-DG method with slivers as described in Section 4.2 is* L^2 *stable provided*

$$\delta \leq \frac{1}{2} \min\left\{\frac{\Delta t}{\Delta x}, 1\right\}.$$

Proof. We recall the construction in Figure 8. We consider again a periodic degenerate geometry where every other interface is replaced by a sliver element. We focus on a triplet of elements $(C^n_{i-1}, S^n_{i-1/2}, C^n_i)$, with $\delta \in (0, 0.5)$. Up to renormalization, the normals to the four inner sides of the sliver have the coordinates reported in Figure 8.

First, we observe that under the assumption $f(q) = q$, the Rusanov-type ALE flux (12) can be simplified into

$$\mathcal{F}(q^-, q^+, \hat{\mathbf{n}}) = \frac{1}{2}\left[(q^+ + q^-)(\hat{\mathbf{n}}_x + \hat{\mathbf{n}}_t) - (q^+ - q^-)|\hat{\mathbf{n}}_x + \hat{\mathbf{n}}_t|\right],$$

meaning that either q^+ or q^- is selected depending on the sign of $(\hat{\mathbf{n}}_x + \hat{\mathbf{n}}_t)$. In particular, given that $\delta \leq \frac{1}{2}\frac{\Delta t}{\Delta x}$, in the subsequent steps the numerical flux will always select q_{i-1} when evaluated along $\Sigma^{n,-}_{i-1/2}$ and $q^n_{i-1/2}$ when evaluated along $\Sigma^{n,+}_{i-1/2}$, resulting in a pure upwind scheme from the left to the right element.

For given continuous spacetime functions w, v and for all meaningful indices i, n, we define for convenience the following pairings:

$$[w,v]_i^n := \int_{\Omega_i^n} w(x,t^n)v(x,t^n)\,\mathrm{d}x, \qquad \{w,v\}_{i-1/2}^n := \int_{t^n}^{t^{n+1}} w(x_{i-1/2},t)v(x_{i-1/2},t)\,\mathrm{d}t,$$
$$\langle w,v\rangle_i^n := \int_{C_i^n} w(x,t)v(x,t)\,\mathrm{d}x\mathrm{d}t, \qquad \langle w,v\rangle_{i-1/2}^n := \int_{S_{i-1/2}^n} w(x,t)v(x,t)\,\mathrm{d}x\mathrm{d}t,$$
$$\{w,v\}_{i-1/2}^{n,\pm} := \int_{\Sigma_{i-1/2}^{n,\pm}} wv(\mathbf{n}_t+\mathbf{n}_x)\,\mathrm{d}s \quad \text{where } \mathbf{n} \text{ is the outer normal to } \partial S_{i-1/2}^n.$$

Consider now (13) on the element C_i^n: by recalling that we obtained each initial status as $u_i^n(x) = q_i^{n-1}(x,t^n)$ for $x \in \Omega_i^n$, plugging in as test function q_i^n itself into (13), we obtain

$$[q_i^n, q_i^n]_i^{n+1} - [q_i^n, q_i^{n-1}]_i^n - \langle \partial_t q_i^n + \partial_x q_i^n, q_i^n\rangle_i^n - \{q_i^n, q_{i-1/2}^n\}_{i-1/2}^{n,+} + \{q_i^n, q_i^n\}_{i+1/2}^n = 0.$$

By using that $\partial_t((q_i^n)^2) = 2q_i^n\partial_t q_i^n$ and $\partial_x((q_i^n)^2) = 2q_i^n\partial_x q_i^n$, and applying the divergence theorem, we get

$$[q_i^n, q_i^n]_i^{n+1} - [q_i^n, q_i^{n-1}]_i^n - \frac{1}{2}[q_i^n, q_i^n]_i^{n+1} + \frac{1}{2}[q_i^n, q_i^n]_i^n + \frac{1}{2}\{q_i^n, q_i^n\}_{i-1/2}^{n,+} - \frac{1}{2}\{q_i^n, q_i^n\}_{i+1/2}^n$$
$$- \{q_i^n, q_{i-1/2}^n\}_{i-1/2}^{n,+} + \{q_i^n, q_i^n\}_{i+1/2}^n = 0.$$

By multiplying by 2, adding and subtracting $[q_i^{n-1}, q_i^{n-1}]_i^n$ and $\{q_{i-1/2}^n, q_{i-1/2}^n\}_{i-1/2}^{n,+}$, and rearranging the terms, we obtain

$$[q_i^n, q_i^n]_i^{n+1} - [q_i^{n-1}, q_i^{n-1}]_i^n + [q_i^n - q_i^{n-1}, q_i^n - q_i^{n-1}]_i^n$$
$$+ \{q_i^n, q_i^n\}_{i+1/2}^n + \{1, (q_i^n - q_{i-1/2}^n)^2\}_{i-1/2}^{n,+} - \{q_{i-1/2}^n, q_{i-1/2}^n\}_{i-1/2}^{n,+} = 0.$$

Given our choice of δ, we can easily see that $\{1, (q_i^n - q_{i-1/2}^n)^2\}_{i-1/2}^{n,+} \geq 0$, hence, from the equality above, we get the inequality

$$[q_i^n, q_i^n]_i^{n+1} - [q_i^{n-1}, q_i^{n-1}]_i^n + \{q_i^n, q_i^n\}_{i+1/2}^n - \{q_{i-1/2}^n, q_{i-1/2}^n\}_{i-1/2}^{n,+} \leq 0. \tag{26}$$

Consider now the sliver element $S_{i-1/2}^n$ and the corresponding relation (23): by using as test function $q_{i-1/2}^n$ itself, we readily obtain

$$-\langle \partial_t q_{i-1/2}^n + \partial_x q_{i-1/2}^n, q_{i-1/2}^n\rangle_{i-1/2}^n + \{q_{i-1/2}^n, q_{i-1/2}^n\}_{i-1/2}^{n,+} + \{q_{i-1/2}^n, q_{i-1}^n\}_{i-1/2}^{n,-} = 0.$$

By applying again the divergence theorem to the first term, multiplying by 2, adding and subtracting $\{q_{i-1}^n, q_{i-1}^n\}_{i-1/2}^{n,-}$ and rearranging the terms, we derive

$$\{q_{i-1/2}^n, q_{i-1/2}^n\}_{i-1/2}^{n,+} - \{1, (q_{i-1/2}^n - q_{i-1}^n)^2\}_{i-1/2}^{n,-} + \{q_{i-1}^n, q_{i-1}^n\}_{i-1/2}^{n,-} = 0.$$

By our choice of δ, we have $\{1, (q_{i-1/2}^n - q_{i-1}^n)^2\}_{i-1/2}^{n,-} \leq 0$, leading to

$$\{q_{i-1/2}^n, q_{i-1/2}^n\}_{i-1/2}^{n,+} + \{q_{i-1}^n, q_{i-1}^n\}_{i-1/2}^{n,-} \leq 0. \tag{27}$$

On the left element C_{i-1}^n, the same derivation done on C_i^n applies, and we obtain

$$[q_{i-1}^n, q_{i-1}^n]_{i-1}^{n+1} - [q_{i-1}^{n-1}, q_{i-1}^{n-1}]_{i-1}^n - \{q_{i-1}^n, q_{i-1}^n\}_{i-1/2}^{n,-} - \{q_{i-2}^n, q_{i-2}^n\}_{i-3/2}^n \leq 0. \tag{28}$$

By taking the sum of (26), (27) and (28), we get

$$[q_i^n, q_i^n]_i^{n+1} - [q_i^{n-1}, q_i^{n-1}]_i^n + [q_{i-1}^n, q_{i-1}^n]_{i-1}^{n+1} - [q_{i-1}^{n-1}, q_{i-1}^{n-1}]_{i-1}^n + \{q_i^n, q_i^n\}_{i+1/2}^n - \{q_{i-2}^n, q_{i-2}^n\}_{i-3/2}^n \leq 0.$$

We sum now over all positive indices i and over time, take into account the periodicity assumption to cancel out boundary terms, and obtain

$$\int_\Omega (q^n(x,t^{n+1}))^2 - (u^0(x))^2\,\mathrm{d}x \leq 0,$$

which concludes the proof. □

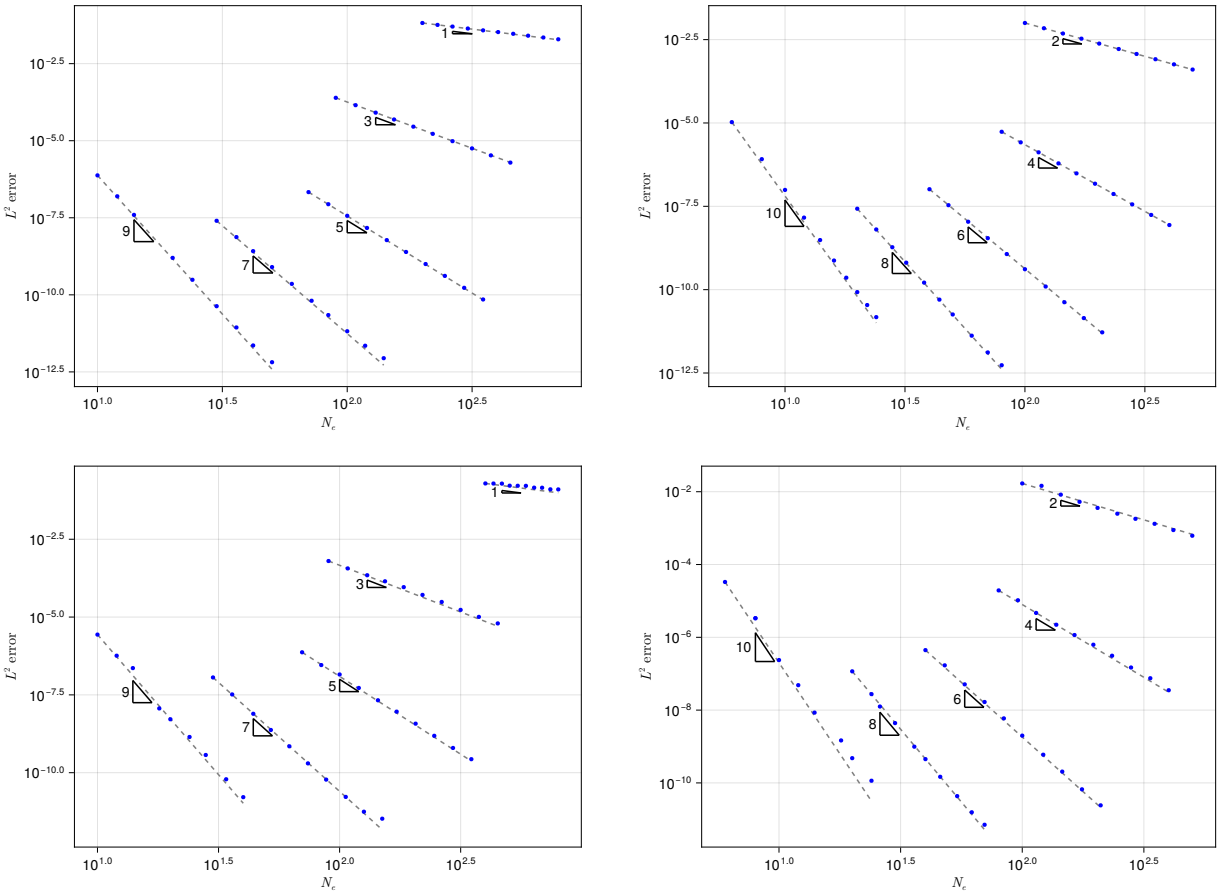


Figure 11: Convergence order of the explicit (top) and the implicit (bottom) ADER-DG method with slivers: left odd orders for $N = 0, 2, 4, 6, 8$, right even orders for $N = 1, 3, 5, 7, 9$.

4.4. Numerical convergence with spacetime hole-like sliver elements

As both the explicit and the implicit ADER-DG methods on degenerate geometries are derived as generalisations of the ones on classical geometries, we expect them to have order of convergence $N + 1$ for a given polynomial degree N. To verify this property we repeat the same numerical analysis of Sections 3.2 and 3.4, over a domain where we insert a sliver element at each interface (except at the boundary), with $\delta = 0.2$. As in the classical setting, we set a CFL at $0.9 \cdot \mathrm{CFL}_{\max}$ for the explicit method and at $10 \cdot \mathrm{CFL}_{\max}$ for the implicit method, where $\mathrm{CFL}_{\max}$ is taken from the second row of Table 2. In Figure 11, we report L^2 errors with respect to the exact solution at the respective final times ($T = 12$ for the explicit and $T = 1$ for the implicit). Each convergence order is correctly achieved as expected.

5. Conclusions and outlook to future works

In this paper, we studied the von Neumann stability and the convergence of the family of ADER-DG methods for $N = 1, \ldots, 9$, considering both explicit and implicit formulations, all within the ALE framework. In particular, we showed that the use of degenerate spacetime geometries leads to some equivalent CFL-type stability bounds as those governing stability in the case of classical geometries. Therefore, elements with zero spatial size do not lead to a reduction of the time step. This result is also important because it provides a theoretical foundation for the use of degenerate elements to connect moving meshes in multiple dimensions, and it represents a natural starting point for the construction of new spacetime cut cell-based methods [49, 50].

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